

Medical Insurance Regression Report

Prediction target and selected model

I used multiple linear regression to predict individual medical charges recorded in the charges column, using age, BMI, number of children, sex, smoking status, and region. I selected the model with a BMI and smoking interaction because five-fold cross-validation on the training records gave it a lower average RMSE, about \$5050 compared with \$6297 for the basic model.

Model performance

- **R²:** The model achieved a test R² of 0.894 on 268 records, indicating a strong overall fit for these test records.
- **RMSE:** Test RMSE was about \$3906, about 67% lower than the \$12013 from giving every test record the training average charge.
- **MAE:** Test MAE fell from about \$9081 for the average prediction to \$2594 for the model, reducing average prediction error by about 71%.

Conclusions and insights

- **BMI and smoking interaction:** Adding the BMI and smoking interaction improved prediction accuracy, reducing average validation RMSE by about 20% compared with the basic multiple linear regression model.
- **Smoking status:** Smoking status is a relevant predictor for this dataset because smokers and nonsmokers have substantially different charge levels.
- **Prediction limits:** Further improvement is needed because the model predicts charges higher than the actual charges for many nonsmokers. However, for some other records, the model also predicts charges lower than the actual charges, sometimes by nearly \$20000.
- **Statistical relationships:** These findings support statistical conclusions about this dataset, but they do not establish that changing a predictor causes a person's actual medical charges to change.